Semantic-Explicit Filtering Network for Remote Sensing Image Change Detection

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***Abstract*— Remote sensing image change detection (RSI-CD) aims to explore surface change information from aligned dual-phase images. However, RSI-CD currently encounters two major challenges. The first issue is the inadequate object-level semantic representation during the feature extraction in CD networks. The other issue is the spectral resolution of the RS image is limited, which leads to a mixture of pseudochange and real change. In order to explore the above-mentioned two challenges, we propose a semantic-explicit filtering network (SFNet) based on a neighborhood feature attention module (NFAM) and multiple-receptive-field semantic filtering mechanism (MSFM). First, the NFAM exploits the correlation of multiscale features and fuses features from the proximity layer to enhance the semantic-explicit representation of the object level. Then, the MSFM takes the weight map after the enhanced semantic representation as input and progressively refines the weight map through a multiple-receptive-field parallel convolution (MPC). This process filters out pseudochange from the predicted result while retaining the real-change information. The experiments on two benchmark datasets demonstrate that the proposed approach presents satisfactory performance over the existing methods.**

***Index Terms*— Change detection (CD), multiple receptive field (RF), neighborhood feature attention, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) is one of the key research topics in remote sensing (RS). The goal of CD is to assign binary labels (i.e., with or without change) to each pixel within the same region by analyzing objects present in that region from images captured at disparate time intervals [1]. In recent years, the rise in demand for high-resolution RS image CD (RSI-CD) has made its applications more important. Therefore, it is greatly significant in many fields, such as urban planning [2], resource management [3], damage assessment [4], [5], and so on.

Received 19 June 2024; revised 6 September 2024; accepted 29 September 2024. Date of publication 9 October 2024; date of current version 28 October 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160401, in part by the National Natural Science Foundation of China under Grant 62301385, and in part by the Key Research and Development Program of Shaanxi under Grant 2024GX-YBXM-130. *(Corresponding author: Qiang Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3476992

Although the rise in demand for RSI-CD offers new opportunities, these opportunities come with great challenges. First, objective factors, such as brightness, weather, and background, at different times create discrepancies, thereby leading to the same object appearing with different colors and shapes in dual-phase images. Second, the object region appears in different sizes, it requires consideration of both small-scale and large-scale changes. Third, the information obtained by traditional CD methods is coarse grained and lacks detailed semantic information about the changes. In many applications, it is crucial to consider both the positional information provided in pixel-level changes and the semantic information contained in object-level changes. Finally, pseudochange interferes with the semantic recognition of real change features [6]. These challenges lead to objects with the same semantic concepts featuring different features.

Currently, deep-learning-based CD methods become the mainstream methods in the field of RSI-CD. It is categorized into supervised [7], [8] and unsupervised methods [9]. Specifically, supervised CD methods [10], [11] primarily rely on deep convolutional neural networks (CNNs) [12]-based frameworks to extract high-level semantic features from dual-phase images. Furthermore, it is crucial to effectively integrate contextual information across different spatial and temporal scales in RSI-CD. To achieve this, researchers improve the abilities of the network to capture a wider range of spatial information by adding multiple convolutional layers [13], [14], [15], utilizing deformable convolution [16], [17], and applying attention mechanism [18], [19]. Additionally, traditional CNN usually employs fixed receptive fields (RFs), while the integration of the attention mechanism into CNN allows for the effective incorporation of global contextual information [20]. For example, Gao et al. [21] propose the attention-based multiscale input–output network, which improves the recognition of small objects by analyzing deep features within the framework of global contextual semantics. Although the method effectively integrates both the CD and attention mechanism, it could be further improved in filtering pseudochange information. Technically, a robust RSI-CD network is considered to be able to focus on object-level semantic changes in the region of interest and mine both local and global features.

To effectively accomplish the RSI-CD task, many CD methods are proposed. However, they still remain faced with the following issues. First, the feature representation capability of the network is limited, and clear object-level semantic features are gradually lost during multilayer feature extraction. Second,

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the CD process is highly sensitive to pseudochange in images, and this overreaction to pseudochange hinders the accurate detection of real changes. To address these issues, this article proposes a novel approach called the semantic-explicit filtering network (SFNet). It is designed to enhance the object-level semantic explicit representation and alleviate the influence of pseudochange, thereby improving the performance of RSI-CD. The main contributions of this work are as follows.

1) A neighborhood feature attention module (NFAM) is proposed. It integrates multiscale features with their corresponding neighborhood features to generate clear object-level semantic features. This module solves the problem of insufficient representation of object-level semantic features.
2) A multiple-receptive-field semantic filtering mechanism (MSFM) is proposed. It performs multiple-receptive-field parallel convolution (MPC) on the decoder output and incorporates residual structure and linear embedding module to filter out pseudochange information. This process preserves the integrity of the real changes.
3) The experiment demonstrates that the proposed NFAM and MSFM are effective for model optimization and learning. Besides, the comparison experiment indicates that the proposed method also effectively detects change information. Extensive experimental results show that the proposed SFNet achieves higher CD accuracy than most popular CD networks.

The rest of this article is organized as follows. Section II analyzes related work, including traditional CD methods, deep-learning-based CD methods, and transformer-based CD methods. Section III analyzes the details of the proposed method. Section IV presents and analyzes the results of the experiment, and Section V arrives at a conclusion.

II. Related Work

In this section, we briefly introduce the current related work on CD from three aspects: traditional CD methods, deep-learning-based CD methods, and attention-mechanism-based CD methods.

A. Traditional CD Methods

Traditional CD methods primarily obtain a change map by comparing pixel information in the same region of dual-phase images. These methods typically include direct comparison, transformation-based, and machine learning (ML)-based methods. Initially, the direct comparison methods use algebraic calculations to set thresholds to achieve naive CD. Common algebraic computational methods, such as regression analysis [22], image difference [23], and image ratio methods [24], are each applicable to discrete dual-phase image inputs. These methods allow direct comparison methods with flexibly respond to a variety of CD needs. Transformation-based CD methods extract features that represent changes in an image by applying some kind of transformation or conversion to the input images. These methods are more adaptable to complex scenes, such as change vector analysis (CVA) [25], principal component analysis (PCA) [26], and texture-based analysis (TBA) methods [27], and so on. The TBA-type CD method is utilized to extract texture features from images, which allows for the identification of nuanced changes in texture. Similarly, the PCA-type CD method is used to identify the components that represent the main features, and this helps to reduce the data dimensions and detect patterns of image changes. Finally, the CVA-type CD method categorizes various types of changes through the analysis of transformation vectors between pixel pairs. These three methods complement one another and are commonly used in conjunction to improve the accuracy of RSI-CD. Although the transformation-based CD method provides more detailed change information compared to direct comparison methods, it is more susceptible to image quality issues, which may lead to significant interference terms in the result. In addition to direct comparison and transformation-based methods, ML-based CD methods, such as random forests [28], artificial neural networks [29], Markov random field [30], and support vector machines [31], are increasingly becoming the mainstream methods for traditional CD. Technologically, the success of artificial neural networks-based CD methods provides the foundation for the subsequent development of deep-learning-based CD methods.

B. Deep-Learning-Based CD Methods

To address the balance between texture and details when extracting object features, researchers propose the theoretical framework based on full CNNs [32]. Currently, the U-Net [33] framework is popular in the field of RSI-CD due to its excellent capabilities in feature extraction and fusion. It is designed for semantic segmentation to extract hierarchical semantic features from dual-temporal images through its encoder to enhance contextual understanding. Subsequently, it classifies the pixel-level information through the decoder to generate an accurate segmentation map. For example, Huang et al. [34] build a Siamese neural network structure based on U-Net to improve the input comparison strategy for images. Fang et al. [35] propose a Siamese network that fuses an integrated channel attention module with nested U-Net [36] to extract deep and shallow features from dual-temporal images. Raza et al. [37] utilize nested U-Net to integrate global and local features, thereby achieving multiscale feature fusion and generating the final change map. Furthermore, U-Net is often used with metric learning in RSI-CD tasks, which is also an important research direction. The essence of metric learning is to generate the change map directly by calculating feature distances, thus bypassing the upsampling step in traditional decoders. For instance, Shi et al. [38] adopt a deep metric learning strategy and incorporate a convolutional block attention module to reduce the impact of pseudochange pixels on CD performance. In current deep learning frameworks, most CD networks face significant challenges in optimal feature extraction and pseudochange information filtering. These aspects are crucial for improving the performance of such networks.

C. Attention Mechanism-Based CD Methods

Attentional mechanisms introduced reasonably in deep learning-based methods can enhance the ability of networks

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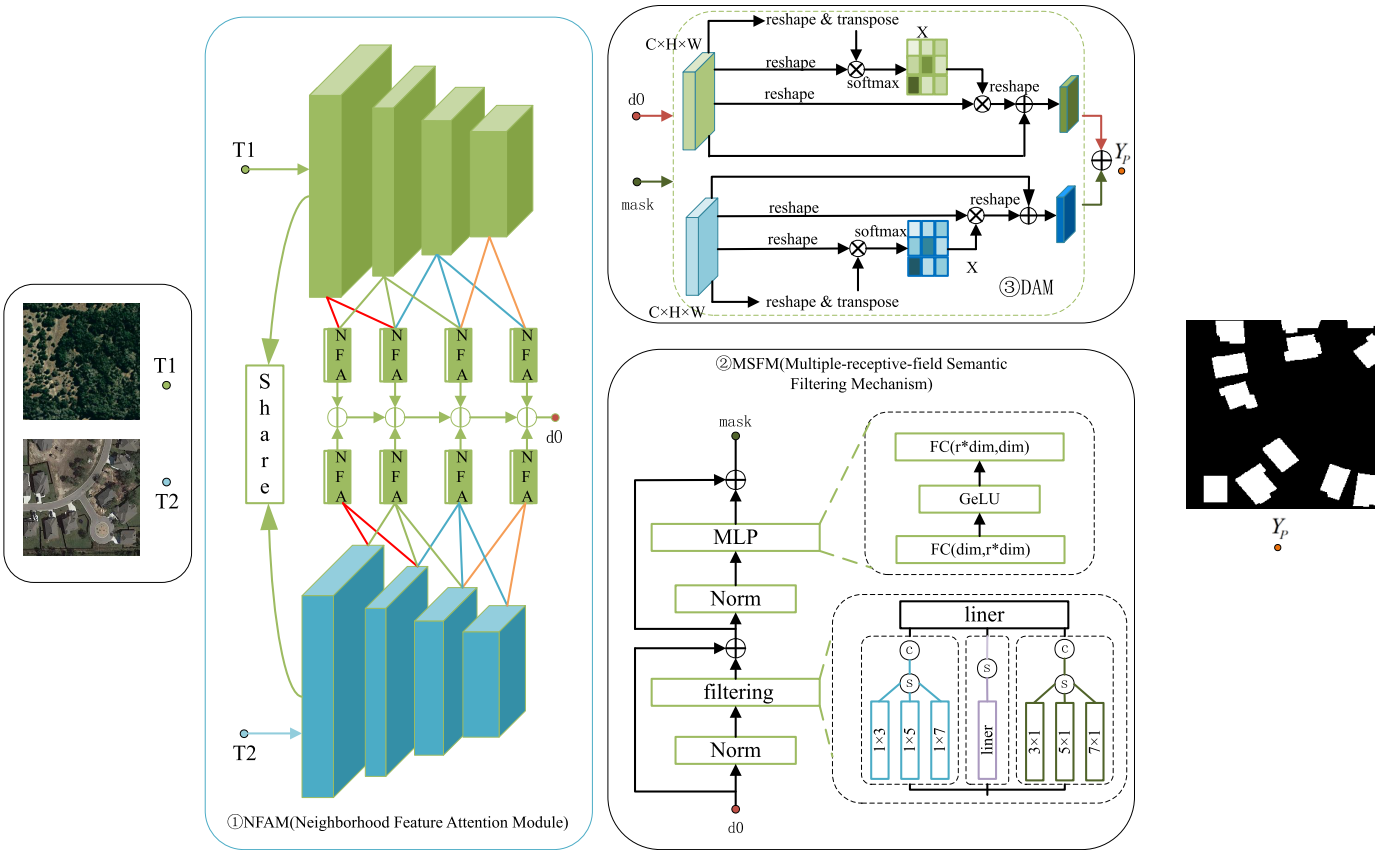


Fig. 1. Overall flowchart of the proposed method. DAM denotes the channel attention mechanism and spatial attention mechanism proposed in DANet to adaptively integrate local features and global dependencies. NFA has the same meaning as NFAM.

to distinguish semantic information. Bahdanau et al. [39] first apply the attention mechanism in machine translation to solve the element's alignment across sequences. Later, the attention mechanism is also extensively used in CD tasks. For example, Chen and Shi [40] combine the concepts of the Siamese network [41] and DANet [42] to design an attention network, which locally associates image spatial features with dimensional features. Lei et al. [43] propose a kernel scale adaptive attention network to learn global features flexibly and reduce the interference of pseudochange. Subsequently, Liang et al. [44] propose a lightweight RS building a CD model, while the self-attention mechanism can break the limitation of RF. Ding et al. [45] propose a multiscale spatial–spectral features aggregation network to solve the dependence on single-scale features for obtaining change information. Furthermore, it is a popular direction in the field of CD to incorporate the attention mechanism in the feature extraction stage. For instance, Peng et al. [46] combine the attention mechanism with nested U-Net to propose a novel generator that realizes RSI-CD within a generative adversarial network (GAN) framework. Huang et al. [47] integrate a selective convolution kernel and attention mechanism into the Siamese network to facilitate feature selection in multiscale dual-phase images. Zhao et al. [48] improve the spatial feature extraction in RSI-CD by carrying out pixel-level spatial correction on multilayer features and filtering out pseudochange information. In contrast to the above-mentioned research, we propose the NFAM module to not only focus on the local and global features but also correlate the features from the neighborhood perspective. Moreover, MSFM can effectively filter pseudochange content and improve the edge of change regions.

III. Proposed Method

In this section, the overall framework proposed is described, and specific explanations are provided for NFAM and MSFM in sequence.

A. Overview

Fig. 1 illustrates the proposed framework. The framework is based on the classical Siamese network and consists of the encoder, NFAM, decoder, MSFM, and DAM. Inspired by adaptively spatial feature fusion (ASFF) [49], NFAM performs neighborhood alignment and adaptive residual fusion of multiscale features, which improves the traditional hierarchical feature fusion and thereby enriches the representation of semantic information. The MSFM optimizes the traditional fixed RF analysis into a multiple-receptive-field fusion analysis, which further reduces the impact of pseudochange in the change region. The specific process of the network framework is shown in the following.

Suppose that dual-phase RSIs $T_1 \in R^{H\times W\times C}$ and $T_2 \in R^{H\times W\times C}$ are used as inputs, where W, H, and C represent the width, height, and number of channels, respectively.

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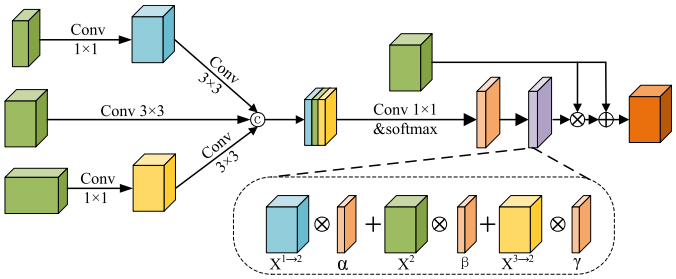


Fig. 2. Schematic of NFAM.

The ground truth (GT) corresponding to these images is $Y \in R^{H\times W\times C}$.

1) In the Siamese framework, the multiscale feature groups $F_1 = \{f_1^1, \ldots, f_1^4\}$ and $F_2 = \{f_2^1, \ldots, f_2^4\}$ of T_1 and T_2 is extracted by the Resnet18 [50] network.
2) The multiscale feature groups F_1 and F_2 are input into NFAM and obtain the context-associated weight maps groups $F_{1N} = \{f_{1N}^1, \ldots, f_{1N}^4\}$ and $F_{2N} = \{f_{2N}^1, \ldots, f_{2N}^4\}$ based on the original feature groups.
3) Calculate the difference between the same layer context-related weight map groups F_{1N} and F_{2N} after NFAM processing. Then, the difference-computed weight maps are decoded together with deeper weight maps to generate a dense feature F_c.
4) Take F_c as input to the MSFM and remove the pseudochange interference of F_c in the MSFM to obtain the real change content F_M.
5) Combine the dense feature F_c and the real change feature F_M by DAM to generate the predicted result Y_p.

B. Neighborhood Feature Attention Module

As the Resnet18 depth increases, the texture information and fine-grained boundaries in the underlying features are progressively abstracted into object localization information [51]. Therefore, the multiscale feature groups F_1 and F_2 generated by the Resnet18 network have different focuses at each layer. Considering that multiscale features have different focuses and higher correlation between neighborhood features [52], we tend to integrate neighborhood features to enhance the representation of extracted temporal features as: 1) extracts semantic and fine-grained information from both low-layer and high-layer features and 2) refines the texture representation of change regions within middle-layer features. The proposed NFAM is shown in Fig. 2.

As shown in Fig. 2, NFAM integrates contextual neighborhood features based on residual structure, and the inputs are object features and its neighborhood features, in order to illustrate the workflow of the NFAM more intuitively. Therefore, the process of f_1^2 is taken as a sample to investigate the relationship between f_1^2 with its neighboring features f_1^1 and f_1^3 in order to generate a weight map for the f_1^2 feature. Among them, f_1^1 is downsampled to match the feature dimension of f_1^2, and its fine-grained information is extracted through a convolutional layer. Similarly, the number of channels in f_1^3 is decreased to match that of f_1^2, and then a convolutional layer is used to extract its semantic information. In contrast, f_1^2 enhances the texture representation of the change region by a convolutional layer and further boosts its expressive ability with a linear mapping layer. Subsequently, the neighborhood features are processed through a weighting operation to generate the augmented result of f_1^2 by incorporating the neighborhood features. Finally, the linear mapping layer is applied to the processed f_1^2, then combined with the enhancement result using a residual structure. This obtains the contextual association weight map f_{1N}^2, which is based on the original features, that is,

$$\text{weight}_n = F_f\left(\alpha f_m^{n-1\to n} + \beta f_m^n + \gamma f_m^{n+1\to n}\right) \tag{1}$$

$$F_{mN} = \left(\text{weight}_n \times f_m^n\right) \times \text{Conv}_{1\times 1}\left(f_m^n\right) \tag{2}$$

where $n = \{1, 2, 3, 4\}$, $m = \{1, 2\}$. If $n - 1 = 0$ or $n + 1 > 4$, it indicates that the feature does not exist. Here, α, β, and γ represent the weights of multiscale neighborhood features, respectively. $F_f(\cdot)$ indicates downsampling and *Softmax* operations, and weight_n denotes the mask content formed after neighborhood feature aggregation. $f_m^{n-1\to n}$ and $f_m^{n+1\to n}$ denote the matching of multiscale neighborhood features to the scale of f_m^n by upsampling and downsampling, respectively. As shown earlier, the NFAM combines the selected features with its neighborhood features and weights them. Therefore, the NFAM can improve the ability of the network to extract change regions and achieve explicit representation of the semantics.

C. Multiple-Receptive-Field Semantic Filtering Mechanism

The NFAM is designed to explore the inherent relationships among neighboring features. It can improve the representation of feature texture and object localization. In contrast, the MSFM aims to decompose and reconstruct the dense feature F_c by MPC. Fig. 1 illustrates the proposed NFAM and the entire process can be depicted as

$$M = M_{\text{atn}}(L_N(F_c)) + F_c \tag{3}$$

$$F_M = M_{LP}(L_N(M)) + M \tag{4}$$

where L_N represents the LayerNorm, and M is a feature within the MSFM. As shown in (3) and (4), the MSFM can be decomposed into two subproblems: multiple-receptive-field integration and spatial location activation.

1) Multiple-Receptive-Field Integration: Since the input feature $F_c \in R^{H\times W\times C_{\text{in}}}$ is a tensor with size $H \times W \times C_{\text{in}}$, it is theoretically feasible for any high-rank matrix to be represented by single rank-deficient vectors. In view of this, we decompose the higher order tensor into single-order vectors by asymmetric convolutional computation in different directions. To reduce parameters and prevent overfitting, our proposed MPC utilizes asymmetric convolution to extract single-rank vectors instead of the traditional point-by-point convolution. Next, the single-rank vectors are nonlinearly mapped using activation functions. Finally, the vectors after mapping are reconstructed cross dimensionally by Flatten processing to generate the higher rank tensor with cleanly varying features. The multiple-receptive-field integration

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is calculated as

$$M_{\text{atn}}(\cdot) = \text{flt}\left(\sum_{i=1}^{C_n} \text{Con}v_{1\times W_i}(\cdot) + \sum_{i=1}^{C_n} \text{Con}v_{H_i\times 1}(\cdot) + \text{Lin}_{C_{\text{in}}\to C_{\text{in}}}(\cdot)\right) \times W^{\text{mlp}}_{c,:} + b \quad (5)$$

where $W_i, H_i = \{3, 5, 7\}$, and $C_n = 3$. $W^{\text{mlp}}_{c,:} \in R^{C_{\text{in}}\times C_{\text{out}}}$ denotes the spatial feature weights computed by the fully connected layer and $b \in R^{C_{\text{out}}}$ is a parameter of $M_{\text{atn}}(\cdot)$. C_n represents the count of MPC. $\text{Lin}_{C_{\text{in}}\to C_{\text{in}}}(\cdot)$ indicates a fully connected computation that matches the number of input feature channels. $\sum_{i=1}^{C_n} \text{Con}v_{1\times W_i}(\cdot)$ denotes to perform MPC, while it uses convolution kernels with sizes 1×3, 1×5, and 1×7 along the horizontal direction, respectively. Similarly, $\sum_{i=1}^{C_n} \text{Con}v_{H_i\times 1}(\cdot)$ represents the filtering in the vertical direction, which employs convolution kernels with sizes 3×1, 5×1, and 7×1, respectively. $\text{flt}(\cdot)$ is the flatten processing.

2) Spatial Location Activation: The feature obtained from the multireceptive-field integration needs to perform spatial location activation to better enhance the filtering effect of pseudochange. It is accomplished by means of a channel scaling operation and an activation function, which both are designed within the residual framework. The spatial position activation is calculated as

$$M_{\text{LP}}(\cdot) = \text{Lin}_{C_{\text{in}}\times r\to C_{\text{in}}}\left(G\left(\text{Lin}_{C_{\text{in}}\to C_{\text{in}}\times r}(\cdot)\right)\right) \quad (6)$$

where r represents the number of channel magnification and $G(\cdot)$ denotes the Gaussian error linear unit (GELU) function. As shown earlier, MSFM achieves pseudochange filtering of features based on the multiple-receptive-field integration and the spatial location activation.

D. Loss Function

To train the network, the binary cross entropy (BCE) loss [53] function is used to measure the discrepancy between predicted result Y_p and GT. To obtain the accurate change predicted result, the model parameters are adjusted iteratively to minimize the BCE loss. The loss is defined as

$$L_{\text{BCE}} = -\frac{1}{H \times W} \sum_{p=1}^{H\times W} \left(Y \log\left(\hat{Y}_p\right) + (1 - Y) \log\left(1 - \hat{Y}_p\right)\right) \quad (7)$$

where Y denotes the label of the pixel. In the predicted result, $\hat{Y}_p$ indicates the probability that the pixel belongs to the change category.

IV. Experiment

A. Datasets

To ensure the generalization capability of the model and the reliability of the experimental results, two open-source datasets are selected as the basis for training and testing after reviewing the relevant literature [54].

1) LEVIR [40]: The dataset contains 637 high-resolution RSI pairs with size 1024×1024 pixels, and it was collected by Google Earth. The dataset covers a period ranging from five to 14 years and captures the change in land use, particularly the expansion of urban areas. It encompasses various types of structures, such as standalone villa houses, multistory apartment buildings, simple garages, large commercial warehouses, and so on.

2) SYSU [38]: The dataset includes 20 000 high-resolution RSI pairs with size 256×256 pixels, and it was collected in Hong Kong from 2007 to 2014. These images offer a detailed view of urban development with a resolution of 0.5 m per pixel. The dataset documents primary changes in the construction of new urban structures, suburban growth, initial infrastructure development, vegetation patterns, road widening, and the reclamation of coastal areas for development projects.

B. Evaluation Metrics

To quantify the effectiveness of the CD, this article utilizes six metrics to evaluate the agreement between the predicted result and the GT. These evaluation metrics encompass precision (Pr), recall (Re), F1-score (F1), overall accuracy (OA), intersection over union (IoU), and kappa.

In the evaluation of model performance, Pr quantifies the ratio of correctly identified positive instances, and a higher Pr indicates a lower false alarm rate. Re represents the percentage of actual samples that are accurately identified, and a higher Re value suggests a lower miss alarm rate. F1 is an indicator that balances Pr and Re to reflect both precision and sample recall. Therefore, the higher F1 scores indicate higher model performance for CD. IoU is an indicator that shows the extent of overlap between the predicted result and GT. OA represents the proportion of samples correctly predicted, and kappa is a refined metric based on OA. The detailed definitions of the above-mentioned evaluation metrics as

$$\text{Pr} = \frac{\text{TP}}{(\text{TP} + \text{FP})} \quad (8)$$

$$\text{Re} = \frac{\text{TP}}{(\text{TP} + \text{FN})} \quad (9)$$

$$\text{F1} = \frac{(2 \times \text{Pr} \times \text{Re})}{(\text{Pr} + \text{Re})} \quad (10)$$

$$\text{IoU} = \frac{\text{TP}}{(\text{FP} + \text{TP} + \text{FN})} \quad (11)$$

$$\text{OA} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})} \quad (12)$$

$$\text{Kappa} = \frac{(\text{OA} - P_e)}{(1 - P_e)} \quad (13)$$

where TP, FN, FP, and TN represent the true positive, false positive, false negative, and true negative, respectively. P_e is an intermediate variable used in the calculation of the kappa, also referred to as the theoretical agreement rate, that is,

$$P_e = \frac{(\text{TP} + \text{TN})(\text{TP} + \text{FP}) + (\text{FN} + \text{FP})(\text{FN} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \quad (14)$$

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TABLE I

RESULTS OF THE QUANTITATIVE COMPARISON OF THE EVALUATION METRICS FOR THE TWO CD TEST SETS. HIGHEST SCORES ARE IN BOLD AND SECOND-HIGHEST SCORE IS UNDERLINED. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%)

Dataset	Method	Pr	Re	F1	OA	IoU	Kappa
LEVIR	STANet	73.80	82.52	77.93	92.22	63.37	71.85
	SemicdNet	90.48	64.86	72.65	93.37	61.07	69.40
	SNUNet	90.28	88.93	89.60	96.86	81.62	87.31
	DSAMNet	89.89	84.89	87.31	96.13	77.61	84.17
	GeSANet	**90.74**	88.76	89.73	96.91	81.32	87.49
	Ours	90.64	**89.59**	**90.11**	**96.96**	**82.25**	**87.98**
SYSU	STANet	40.90	68.89	50.47	69.50	33.75	30.14
	SemicdNet	79.62	60.73	68.91	87.07	52.56	60.93
	SNUNet	63.68	61.30	62.46	82.63	45.42	51.17
	DSAMNet	78.06	60.90	68.42	86.74	52.00	60.19
	GeSANet	79.08	72.09	75.43	88.92	60.55	68.29
	Ours	**80.52**	**75.54**	**77.97**	**89.92**	**63.87**	**71.43**

C. Comparison Algorithms

To ensure objectivity and fairness, this article selects five open-source RSI-CD networks based on the attention mechanism to be compared with SFNet. The specific details of these algorithms are given as follows.

STANet (2020) [40]: The network integrates a pyramid structure with a spatiotemporal attention mechanism to capture the global relationships among pixels across spatiotemporal.

SemicdNet (2021) [46]: The Nested U-Net with an attentional mechanism is proposed as a generator, which can generate fine-grained change maps through the GAN framework.

SNUNet (2022) [35]: The Siamese network, nested U-Net, and ECAM are combined to solve the fuzzy boundary due to multilayer downsampling.

DSAMNet (2022) [38]: Incorporate channel-space dual attention mechanism in each layer of the feature extraction network to generate richer predictions.

GeSANet (2023) [48]: In order to improve geospatial CD accuracy, an RSI-CD network with the ability to correct spatial mismatches and filter pseudochange information is proposed.

D. Implementation Details

Experiments are completed on a platform with the PyTorch 2.0 framework. In the training and testing phase, our network is performed on a server with NVIDIA 1080Ti GPUs. The SYSU dataset satisfies experimental criteria without cropping. In contrast, due to the substantial dimensions of the LEVIR dataset, its training images are subdivided into nonoverlapping images with 256×256. A pretrained ResNet18 model is used as the backbone. The Adam optimizer [55] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is chosen to optimize the network. Following previous works, the learning rate is fixed at 0.001, and its value is halved every 20 epochs.

E. Experimental Results

1) LEVIR Dataset: As shown in Table I, Pr and Re of SemicdNet are lower than SFNet. However, the discrepancy in Re between SemicdNet and our method is more obvious. Therefore, we contend that SemicdNet has limitations in the identification and filtering of pseudochange information, which results in higher miss alarm rates. The Re of STANet significantly exceeds its Pr. In our opinion, the overfiltering of images by the network leads to this phenomenon, which increases the false alarm rate. By comparison, SNUNet, DSAMNet, and GeSANet exhibit relatively stable performance. Notably, GeSANet demonstrates the necessity of exploring feature correlations and pseudochange filtering among multiscale features. The outcomes of quantitative evaluation show that our network obtains the highest Re, F1, OA, and IoU, which are 0.66%, 0.38%, 0.05%, 0.63%, and 0.49% higher than the other methods, respectively.

The images in the fourth and fifth rows of Fig. 3 contain complex changes, such as changes in roads, light, vegetation, and seasons. Due to the insufficient feature extraction and pseudochange filtering ability of the comparison methods, there are obvious pseudochange interference and missed detection in the fourth row of Fig. 3(d)–(h) and the fifth row in Fig. 3(d) and (e), respectively. In contrast, our network predicts content that is closer to the GT when faced with complex scenarios. Therefore, the efficacy of the MSFM in filtering pseudochanges and the capability of the NFAM in representing semantic information are validated.

2) SYSU Dataset: As shown in Table I, the SFNet outperforms the other comparison algorithms on the SYSU dataset. The GeSANet maintains robust network performance in complex scenarios, and its performance is second only to ours in comparison algorithms. This experimental result further highlights the significance of investigating feature correlation and the application of pseudochange filtering within the multiscale feature exploration. In comparison with the LEVIR dataset, SemicdNet exhibits high Pr and demonstrates relatively consistent network performance within the SYSU dataset. In our opinion, the generalization ability of SemicdNet is fully developed in complex scenarios, which makes its network performance relatively stable and reduces the missed alarm rate. Next, STANet and DSAMNet have not exhibited robust performance, as indicated by the large discrepancy between the Re and Pr. This discrepancy is attributed to the overfiltering of networks, which leads to missed detection within the detection results in complex scenarios. Although

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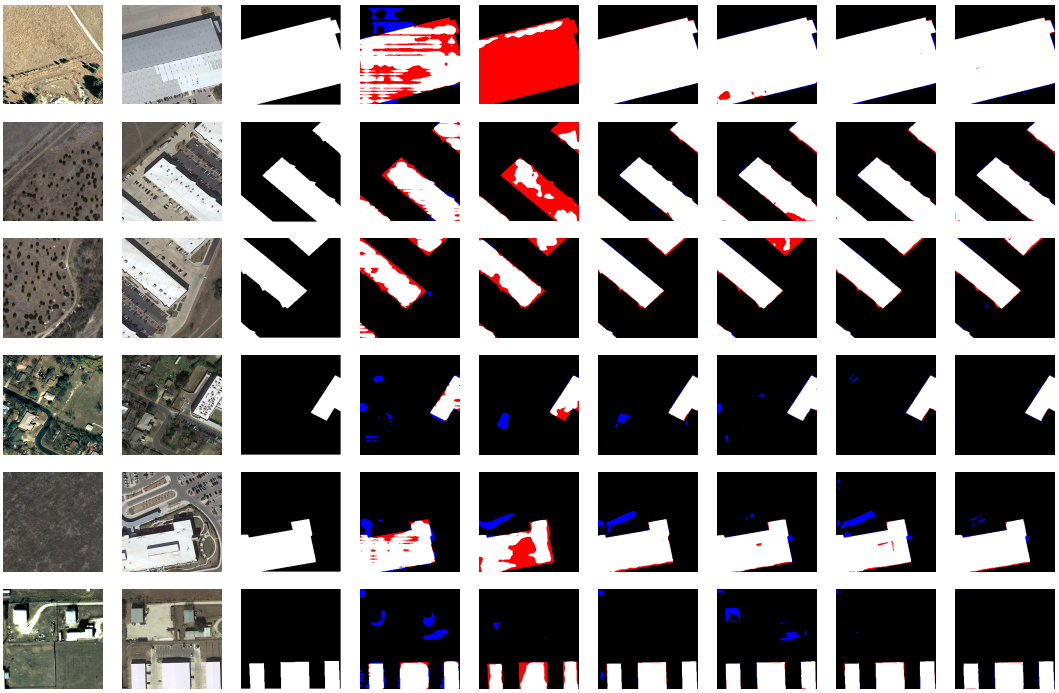


Fig. 3. Visualization results of different methods on the LEVIR dataset. (a) and (b) Original dual-time-phase images. (c) GT. (d) STANet. (e) SemicdNet. (f) SNUNet. (g) DSAMNet. (h) GeSANet. (i) Ours. Blue signifies the pseudochange, red indicates miss detection, and white denotes correct detection.

the SNUNet has no performance imbalance, it still falls short of expectations. Moreover, the complexity of the scenario also affects the overall performance of the comparison algorithms.

In contrast to Fig. 3, the changes included in Fig. 4 are primarily in land, roads, and seasons. It is shown in Fig. 4 that all methods except SFNet have more obvious pseudochange interference and missed detection in the test samples. In the third row of Fig. 4(f)–(h), the prediction results from the three compared algorithms indicate a sequential reduction in missed detections and pseudochange interference. The SNUNet performs the worst due to its lack of the pseudochange filter module. In contrast, DSAMNet and GeSANet with the pseudochange filtering module have relatively few pseudochange in their prediction results. Compared with other methods, our proposed network is also closer to GT in the prediction result of land CD. Therefore, the effectiveness of SFNet in semantic-explicit representation and pseudochange filtering is fully demonstrated.

3) Feature Visualization: To demonstrate the effectiveness of NFAM, Fig. 5 visualizes its results. From Fig. 5, the extracted features can be focused on the change in semantic concepts belonging to the region of interest. Furthermore, feature maps at different scales from the same input show varying levels of texture and structural information. Unlike the traditional backbone to extract features, Scale_2 in Fig. 5 show the results of middle-layer features refining its texture representations after extracting semantic and fine-grained information from low- and high-level layers. Therefore, the earlier validates the ability of NFAM to guide the feature extraction process.

F. Ablation Study

To verify the effectiveness of each component in SFNet, the corresponding ablation experiments are performed in this section. As shown in Table II, the evaluation metrics show how much each component contributes to the performance improvement. It is noted that our proposed network improves significantly on Re, F1, and kappa with baseline. The results of the experiment are discussed in detail in the following four areas.

1) Study of Baseline: In this experiment, DAM is not the main innovation of the article. Therefore, the baseline refers to the original network that contains the DAM. Table II indicates that MSFM exhibits limited improvement over the baseline of the LEVIR dataset. In contrast, the NFAM significantly improves the overall performance of the baseline. This result indicates that NFAM plays a key role in this experiment. In particular, the integration of neighborhood features during the feature extraction phase significantly improves the Pr metric. On the SYSU dataset, MSFM solves the issue of imbalance between Pr and Re in the baseline. It confirms that MSFM is effective in filtering out pseudochange and generating real change maps in scenarios with extremely complex detection backgrounds.

2) Study of NFAM: Since ASFF and NFAM have similar approaches, it is used as a comparison method for NFAM. However, ASFF differs from NFAM in that it assigns equal weight to each feature, thereby ignoring the correlation between neighboring features and introducing additional noise. Therefore, ASFF leads to a decrease in the sensitivity of the backbone network to semantic information. As an example in Table II, ASFF shows weaker performance on both datasets

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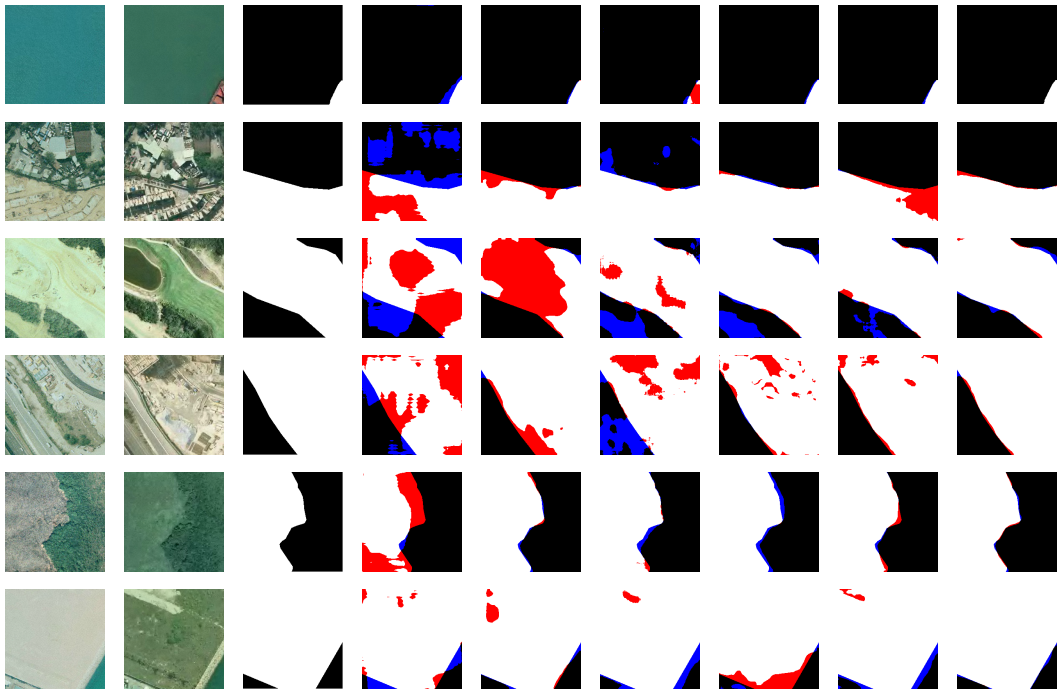


Fig. 4. Visualization results of different methods on the SYSU dataset. (a) and (b) Original dual-time-phase images. (c) GT. (d) STANet. (e) SemicdNet. (f) SNUNet. (g) DSAMNet. (h) GeSANet. (i) Ours. Blue signifies the pseudochange, red indicates miss detection, and white denotes correct detection.

TABLE II

EVALUATION METRICS COMPARISON OF ABLATION EXPERIMENTS. HIGHEST SCORES ARE IN BOLD AND SECOND-HIGHEST SCORE IS UNDERLINED. THE USE OF MODULES OR NOT IS REPRESENTED BY ✓ AND ×. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%)

Dataset	Group	Baseline	NFAM	ASFF	MSFM	Cyclemlp	Pr	Re	F1	OA	IoU	Kappa
LEVIR	Baseline	✓	×	×	×	×	90.44	88.51	89.46	96.81	81.26	87.17
	NFAM	✓	✓	×	×	×	**91.28**	88.91	90.07	96.94	82.23	87.93
	ASFF	✓	×	✓	×	×	89.78	86.21	87.59	96.33	78.73	85.30
	MSFM	✓	×	×	✓	×	90.61	89.04	89.81	96.90	81.86	87.64
	Cyclemlp	✓	×	×	×	✓	89.85	89.02	89.13	96.78	81.25	87.12
	ConNet	✓	×	✓	×	✓	90.18	84.26	86.70	96.10	77.57	84.28
	Ours(ResNet34)	✓	✓	×	✓	×	90.76	89.41	90.08	96.93	82.23	87.95
	Ours	✓	✓	×	✓	×	90.64	**89.59**	**90.11**	**96.96**	**82.25**	**87.98**
SYSU	Baseline	✓	×	×	×	×	80.20	72.68	76.25	88.51	62.03	69.83
	NFAM	✓	✓	×	×	×	81.01	74.13	77.41	89.88	63.59	71.11
	ASFF	✓	×	✓	×	×	78.88	73.25	75.96	89.07	61.24	68.90
	MSFM	✓	×	×	✓	×	79.85	75.05	77.37	89.77	63.15	70.84
	Cyclemlp	✓	×	×	×	✓	23.28	50.78	31.92	53.24	18.99	13.29
	ConNet	✓	×	✓	×	✓	70.98	70.24	70.61	86.21	54.57	61.60
	Ours(ResNet34)	✓	✓	×	✓	×	**81.13**	75.33	77.67	89.90	63.64	71.21
	Ours	✓	✓	×	✓	×	80.52	**75.54**	**77.97**	**89.92**	**63.87**	**71.43**

compared with NFAM. This demonstrates the effectiveness of the integration of neighborhood features by NFAM to enhance the semantic-explicit representation.

3) Study of MSFM: Since Cyclemlp [56] is similar to the basic concept of MSFM, it is compared with MSFM by referring to the ablation experiment of NFAM. However, Cyclemlp utilizes a cross-shaped structure for feature extraction, which fixes RF. This design reduces the ability of the network to filter pseudochange. Table II shows that the pseudochange filtering ability of Cyclemlp is close to MSFM in a building CD dataset LEVIR. In contrast, this narrow RF analysis on the SYSU dataset is insufficient for complex scenarios. Moreover, this insufficiency leads to discrepancies between MSFM and Cyclemlp in terms of CD performance.

4) Study of Module Complementarity: To validate the complementary nature of NFAM and MSFM, we combine ASFF with Cyclemlp to generate a new network (ConNet). As shown in Table II, ConNet is inferior to ASFF in all metrics except for Pr on the LEVIR dataset. In our opinion, the feature weight assignment approach of ASFF that ignores the correlation between neighboring features introduces additional noise, and the feature fusion approach of Cyclemlp enhances the interference of this noise. Moreover, it fails to strictly filter pseudochange when dealing with the CD task in complex

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TABLE III
ABLATION STUDY OF SFNET WHEN FUSING DIFFERENT TIMESTAMP FEATURES. HIGHEST SCORES ARE IN BOLD AND SECOND-HIGHEST SCORE IS UNDERLINED. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%)

Dataset	Group	Pr	Re	F1	OA	IoU	Kappa
LEVIR	Ours(addition and concatenation)	88.71	88.96	88.40	96.59	80.28	86.30
	Ours	**90.64**	**89.59**	**90.11**	**96.96**	**82.25**	**87.98**
SYSU	Ours(addition and concatenation)	75.65	**79.56**	77.55	89.14	63.34	70.40
	Ours	**80.52**	<u>75.54</u>	**77.97**	**89.92**	**63.87**	**71.43**

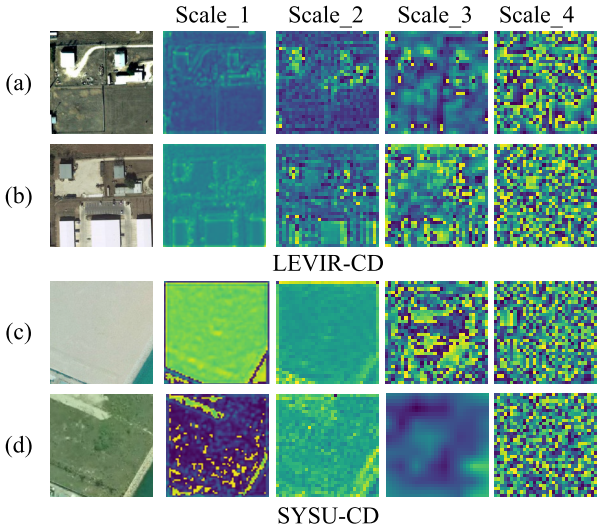


Fig. 5. Feature visualization of the proposed NFAM. (a) and (b) Visualization for dual-phase images on the LEVIR. (c) and (d) Visualization for dual-phase images on the SYSU.

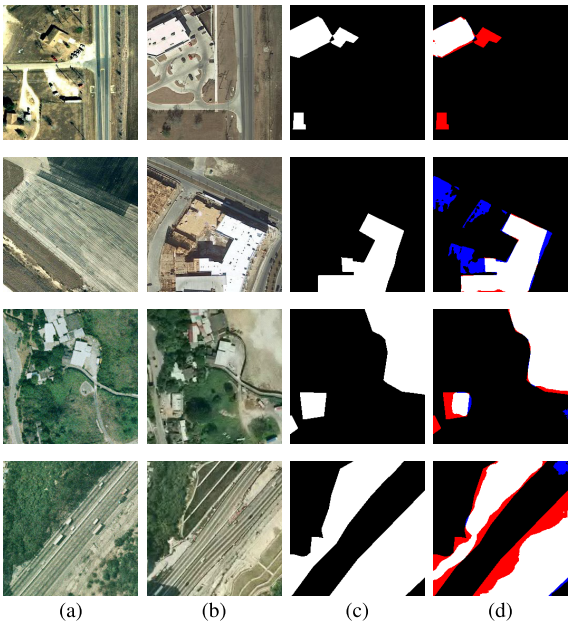


Fig. 6. Visualization of the limitations in SFNet across various datasets: (a) and (b) Original dual-time-phase images. (c) GT. (d) Ours. Blue signifies the pseudochange, red indicates miss detection, and white denotes correct detection.

scenarios, which makes the CD effect of ASFF disturbed. Compared with the metrics of the other submodules, NFAM exhibits high Pr and MSFM exhibits high Re, and SFNet achieves comparable performance with the two modules combined. Therefore, the complementarity between NFAM and MSFM in SFNet is validated.

5) Study of SFNet: To further evaluate the SFNet, we modify the backbone and feature fusion approach while maintaining the SFNet architecture. As shown in Table II, the evaluation metrics of ResNet34 are suboptimal, which may be due to its greater ease of overfitting with the same parameters, and thus ResNet18 is more suitable as a backbone for SFNet. Compared with subtraction, addition and connection preserve more features; however, they are not as focused on detecting change as subtraction. As shown in Table III, SFNet is more likely to achieve comparable performance when using subtraction to fuse features of different time stamps.

In summary, SFNet integrates the advantages of NFAM andMSFM so that the two modules work together to achieve complementary effects. This integration makes SFNet effective in filtering pseudochange due to seasonal effects, shadow casting, and so on, thus enhancing the semantic-explicit representation of features.

G. Limitation

The improvements that are discussed in Section IV-F confirm that SFNet has better CD performance compared with other comparison algorithms. Although SFNet explores the challenges of semantic-explicit representation and pseudochange interference in the CD process, it still has shortcomings that can be improved. In Fig. 6, we select two representative results from the LEVIR and SYSU datasets, respectively. The first two rows of Fig. 6 show that SFNet predicts building changes while also being affected by land changes. Specifically, the risk of misclassification rises when the color of the land resembles that of the buildings. Although NFAM enhances the sensitivity of the network to semantic information, this enhancement needs to be optimized for simple background CD tasks. In the last two rows of Fig. 6, the prediction results of SFNet deviate from the human assessment due to the difficulty in distinguishing the change region from the background. Although MSFM is designed to filter out pseudochange, it is still insufficient for cases where the change region is extremely similar to the background. Therefore, the performance of NFAM and MSFM can be further improved.

V. CONCLUSION

This article proposes an SFNet for RSI-CD. It incorporates an NFAM and an MSFM to address the challenges

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of semantic-explicit representation and the inadequate suppression of pseudochange during the RSI-CD process. The neighborhood features attention module exploits the correlation of multiscale features in the backbone and integrates the semantic information of neighboring features to generate masks. Then, it employs these masks to weigh the multiscale features, thereby enhancing semantic-explicit representation. Additionally, MSFM improves the detection of the network further by filtering the pseudochange pixels through the MPC. Experiments show that both modules are effective in performance improvement. Furthermore, results on two datasets demonstrate that our network can achieve comparable performance. In the future, we will increase the dropout probability and constrain it with a loss function during feature extraction to improve the generalization ability of the network.

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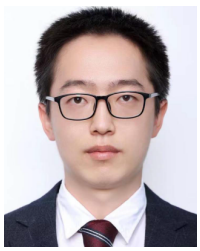
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